

Projection

Introduction



Projection

✓ Is a type of Transformation

$$m = n - 1$$

✓ Transforming points in a coordinate of dimension n into a coordinate system of dimension m where $m < n$

{ **Planar Geometric Projection:** Projecting an object on to a flat plane

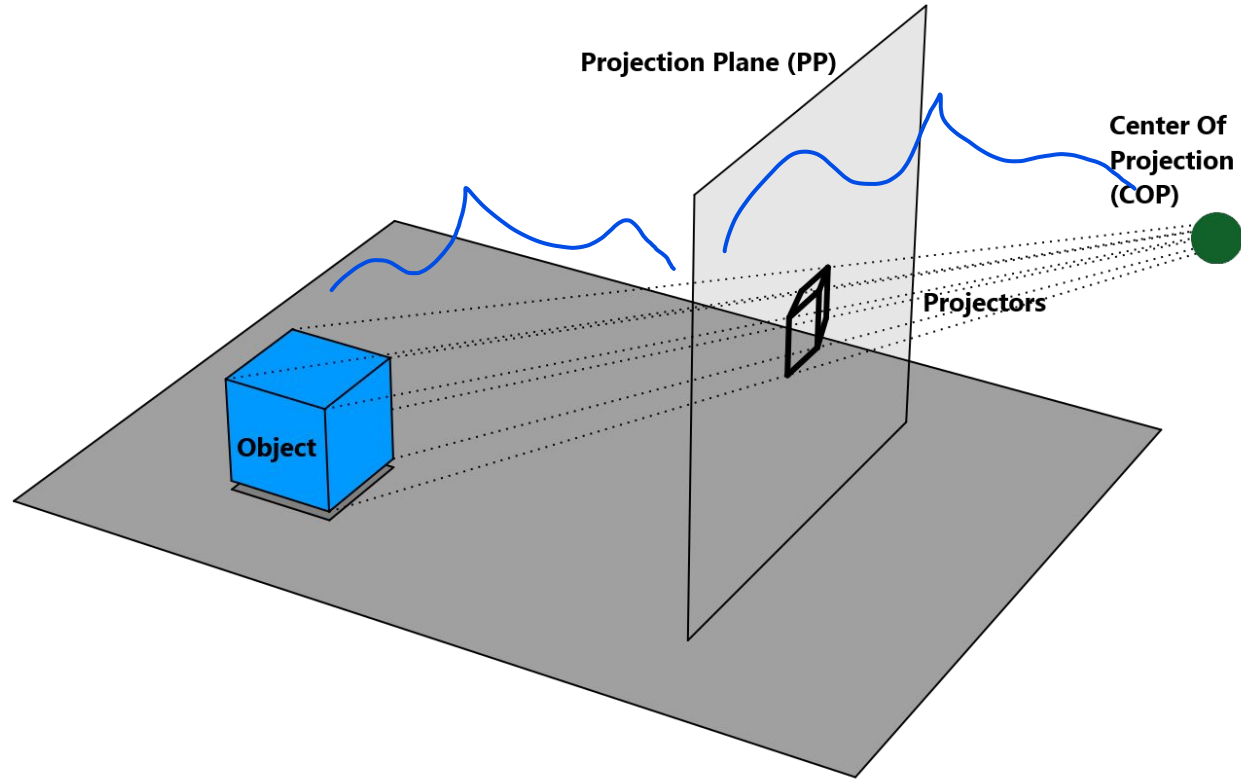


- ✓ ● Turning 3D shapes to its 2D plane (this chapter)
- ✓ ● From 2D to 1D also possible

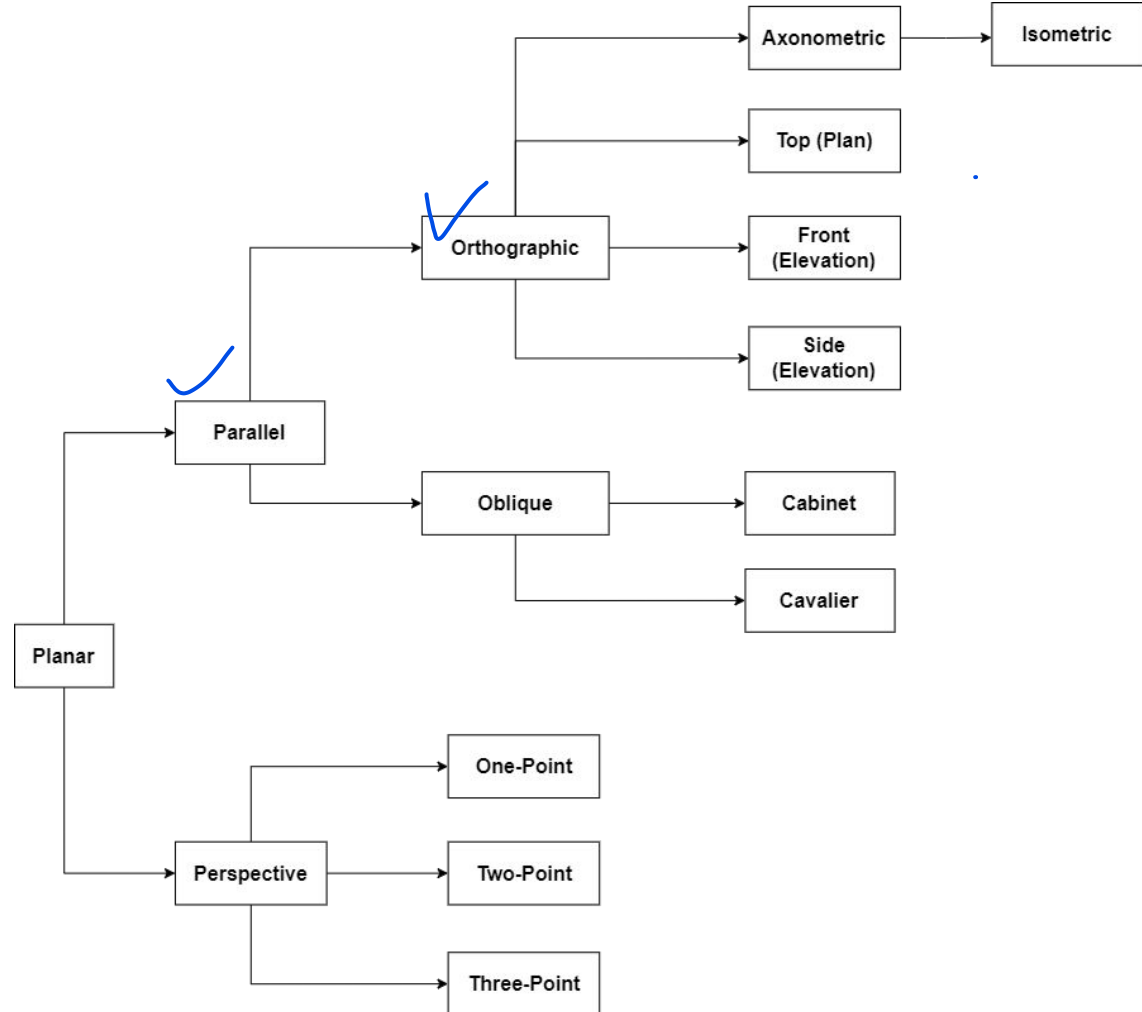
Projection (Terminologies)

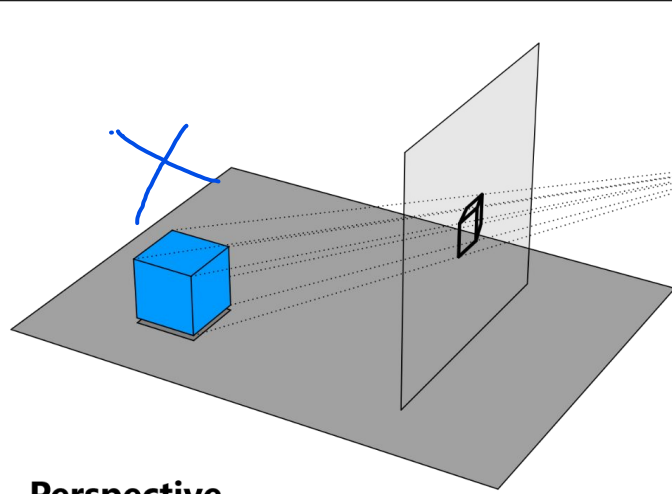
- ✓ • **Object/Target Point (P)**: A single point on an object/geometry we're trying to project (calculate projection of)
- ✓ • **Projected Point (P')**: The projected point after a projection transformation is applied for a given point P'. Projected points together constitutes the projected image
- **Projectors**: The "rays" responsible for projection. They'll intersect the projection plane.
- **Projection Plane (PP)**: The plane where the projected image of the object will appear on. This plane holds all of the P' points. Also known as view plane.
- **Center Of Projection (COP)**: The singular point where all the projectors/projection rays converge (only applicable to perspective projection).
- **Direction Of Projection (DOP)**: The direction towards which the parallel projection rays go towards (only applicable to parallel projection).

Projection (Anatomy)

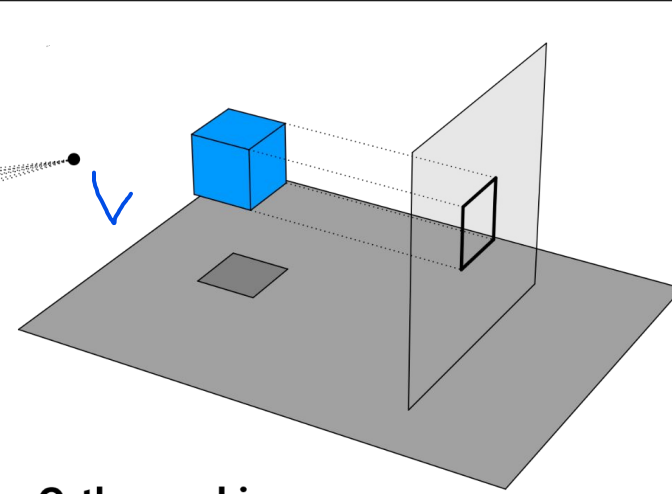


Projection (Types)

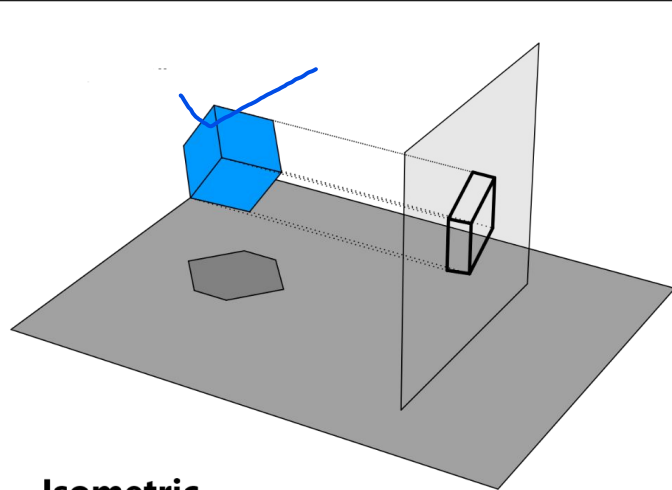




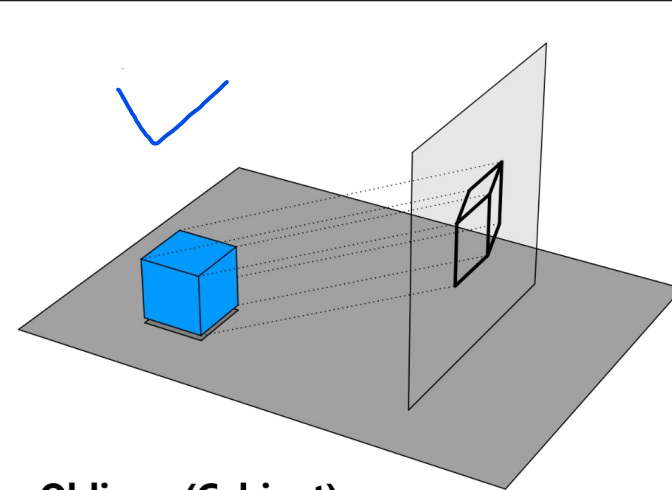
Perspective



Orthographic

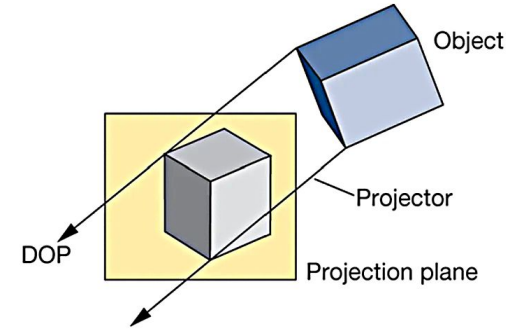
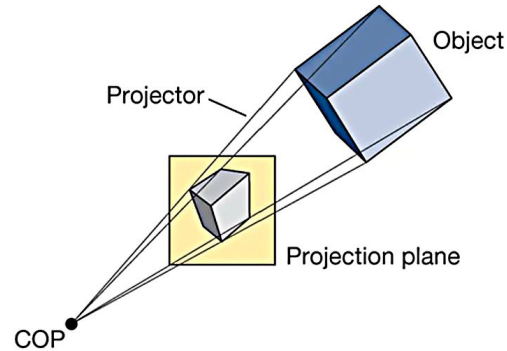


Isometric



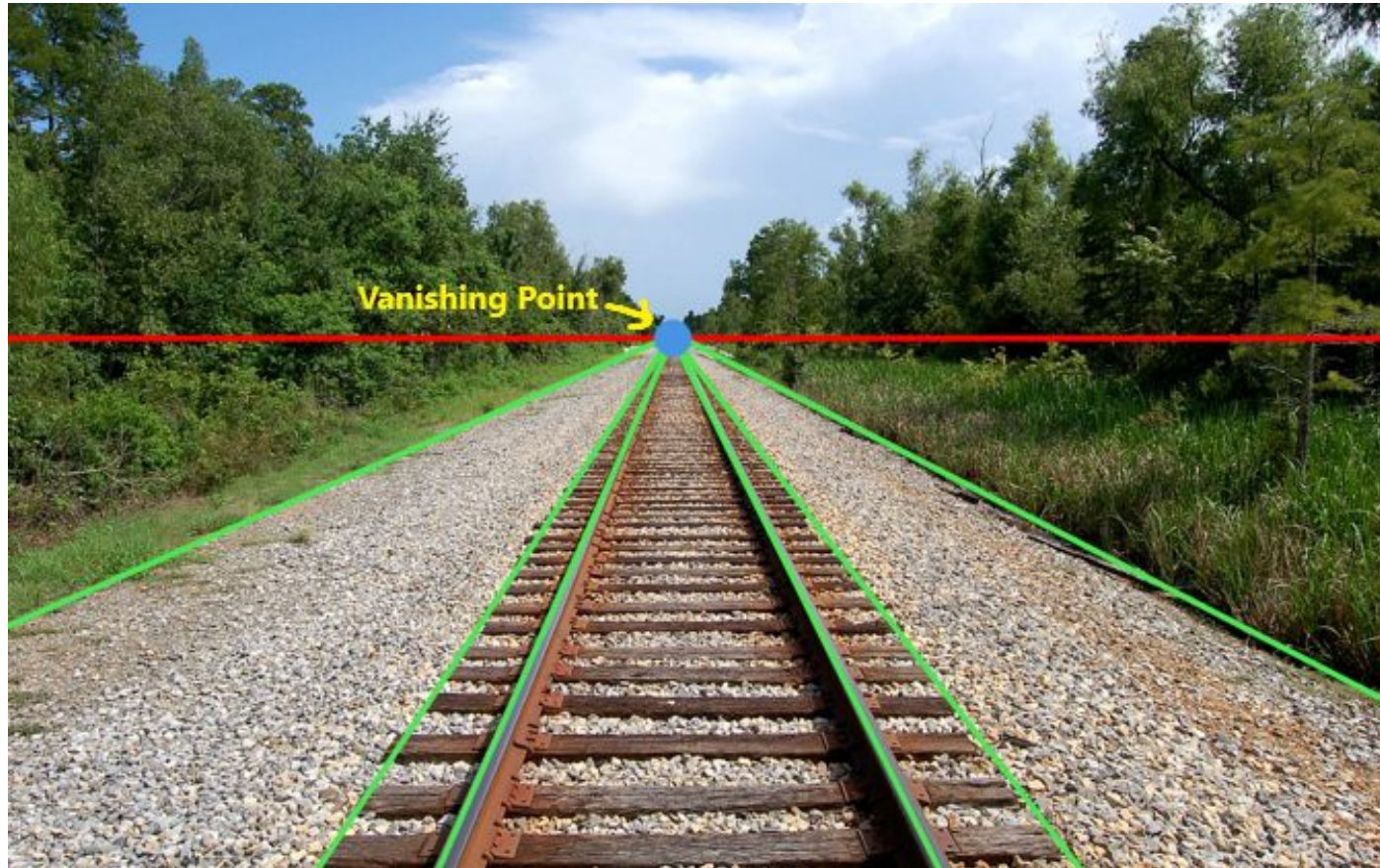
Oblique (Cabinet)

Perspective vs Parallel



	Perspective	Parallel
Nature of Projectors	Projectors aren't parallel, and will converge at a point called COP.	The projectors are always parallel, will never converge.
Configured By	The Center of Projection (COP) and the distance between COP and the projection plane (PP)	The angles those eventually control the Direction of Projection (DOP)
Foreshortening	Yes, so more familiar to human eye	No, appears less realistic
Examples	1-point, 2-point, 3-point	Cabinet, Cavalier, Orthographic

Perspective Projection



Perspective Projection (Attributes)

Vanishing Point: All lines that are parallel to each other (in 3D space) appear to converge into a single point (in projected space)

- One-point Perspective Projection: 1 vanishing point
- Two-point Perspective Projection: 2 vanishing points
- Three-point Perspective Projection: 3 vanishing points

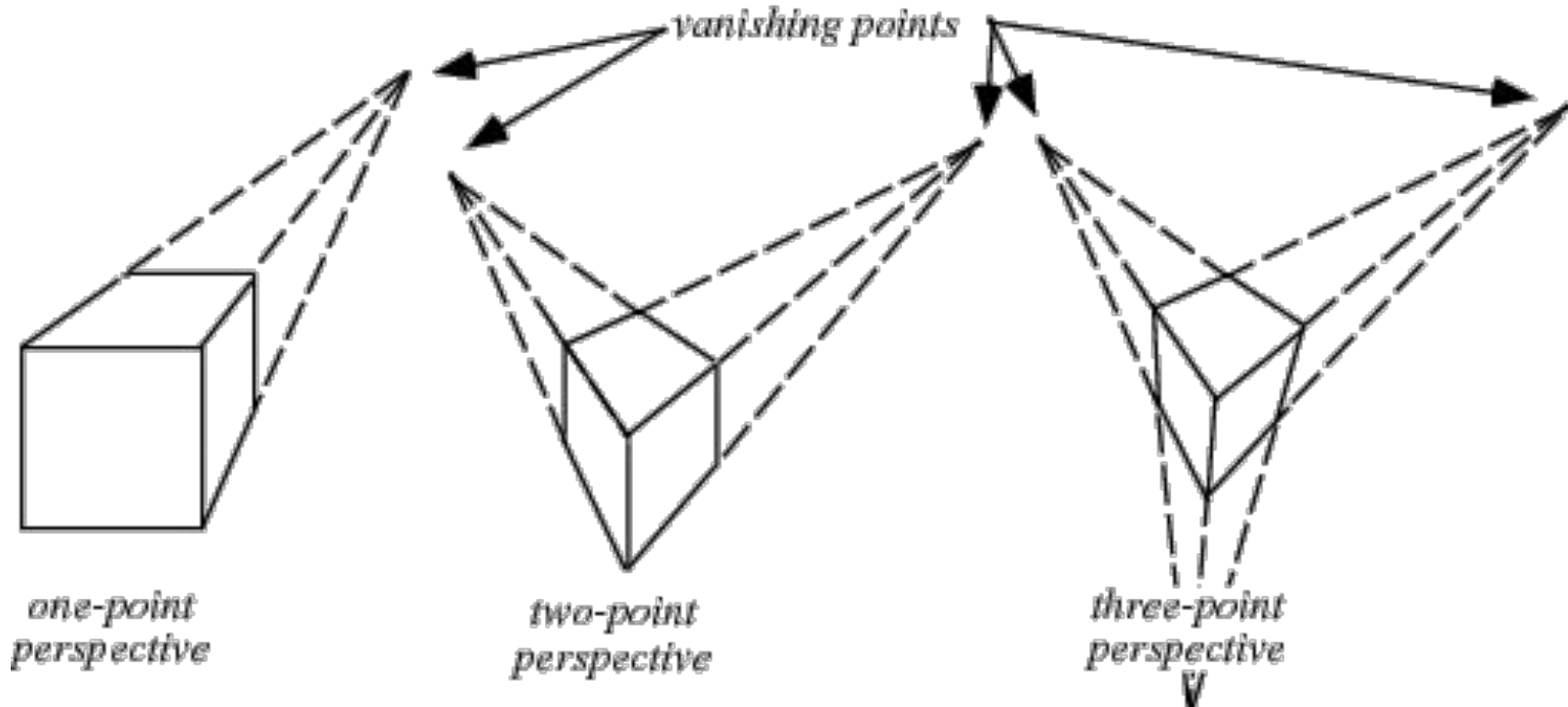
In reality, there are infinite number of vanishing points.

Foreshortening: Basically the phenomenon of far-away objects appearing smaller than the closer ones (despite having equal size).

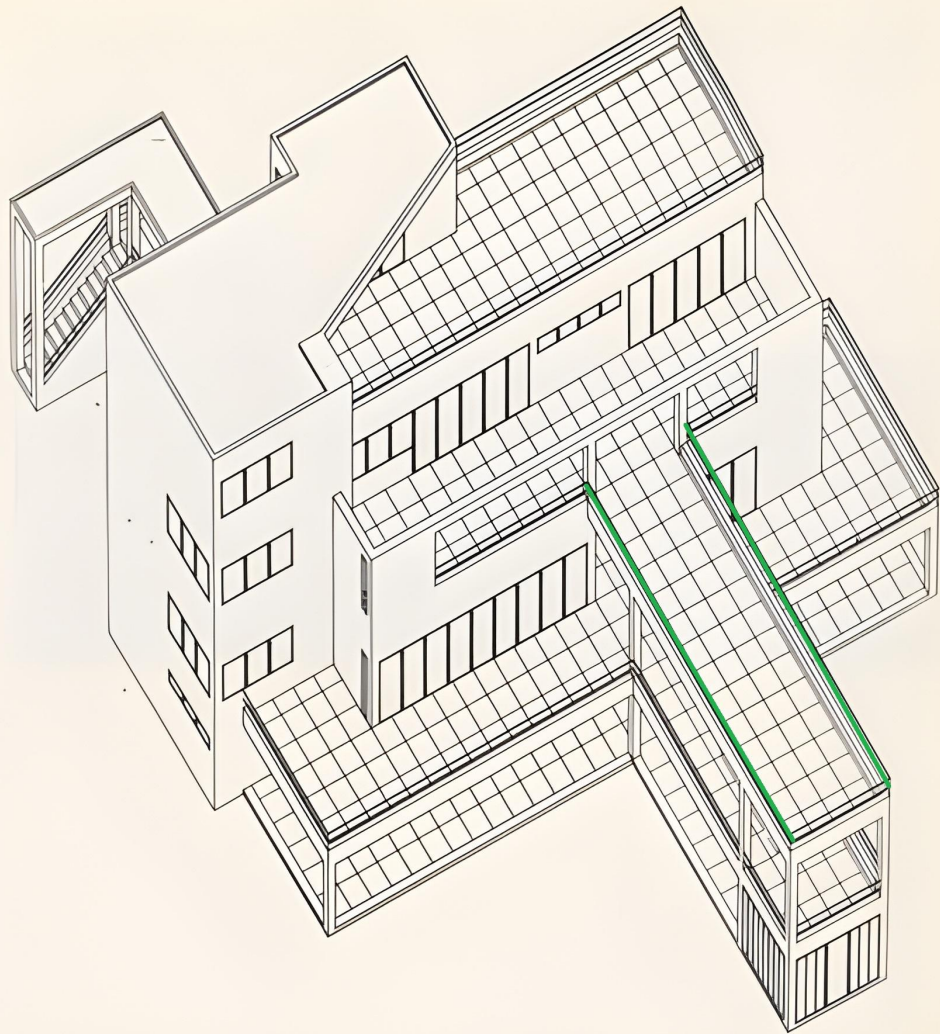
More realistic since resembles human vision. Used in video games, animation, VR



Perspective Projection (Types)



Parallel Projection



Parallel Projection (Attributes)

Parallel Lines: Lines that are parallel to each other (in 3D space) even still appear parallel to each other (in projected space)

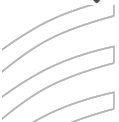
No Convergence: Since the lines remain parallel there are no single point the projectors converge into (in other words, they converge at infinity). So there are no COP, but there is a Direction Of Projection (DOP).

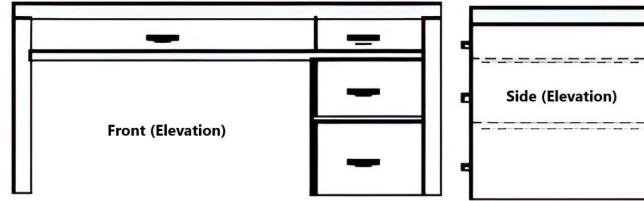
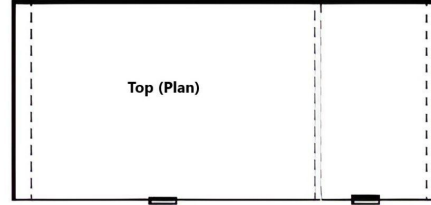
No Foreshortening: Objects that are far-away appear the same size as the closer ones.

For Orthographic the DOP is equal to the normal vector of the plane, whereas they are not equal in case of Oblique.

Parallel Projected Images are less realistic, but they are good for architectural planning and design.

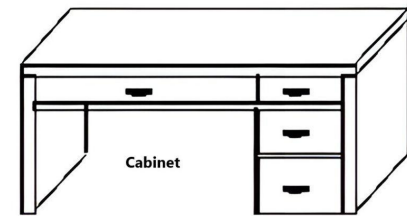
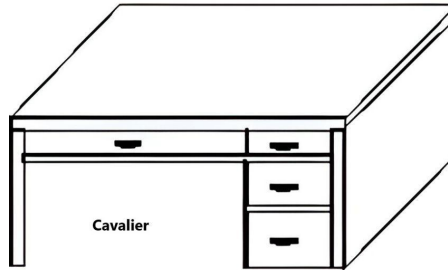
(Some video games use isometric projection which is a type of parallel projection)





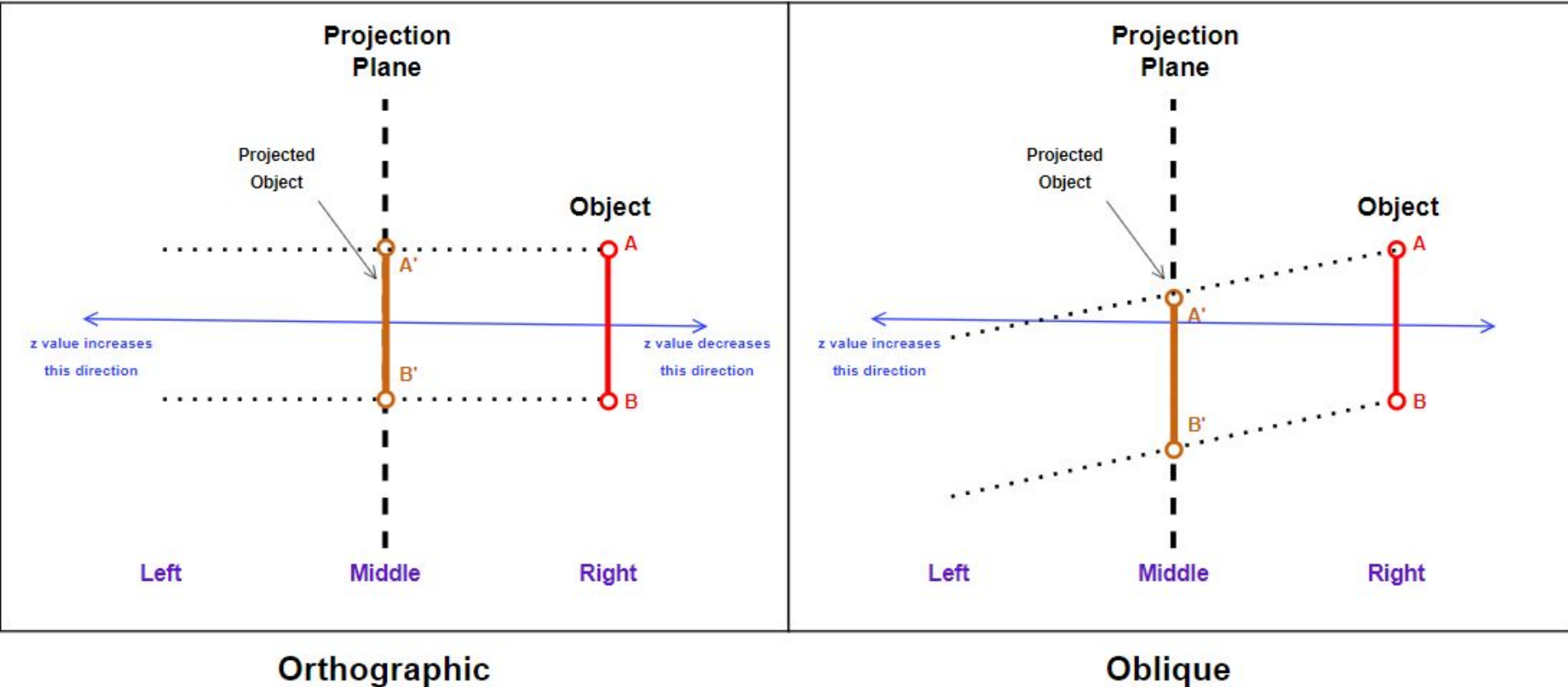
Orthographic

Oblique



Parallel Projection (Types)

Parallel Projection (Geometric Setup)

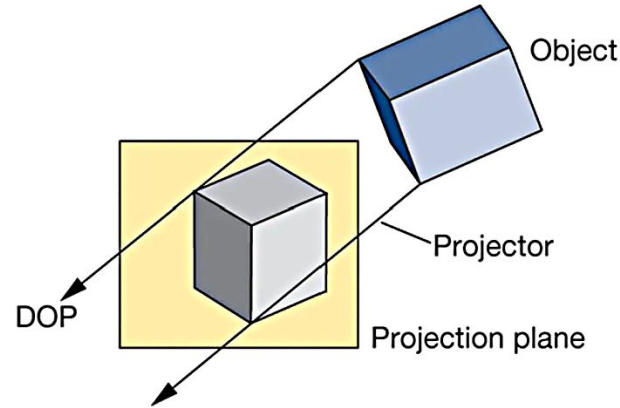


Performing Projection Math

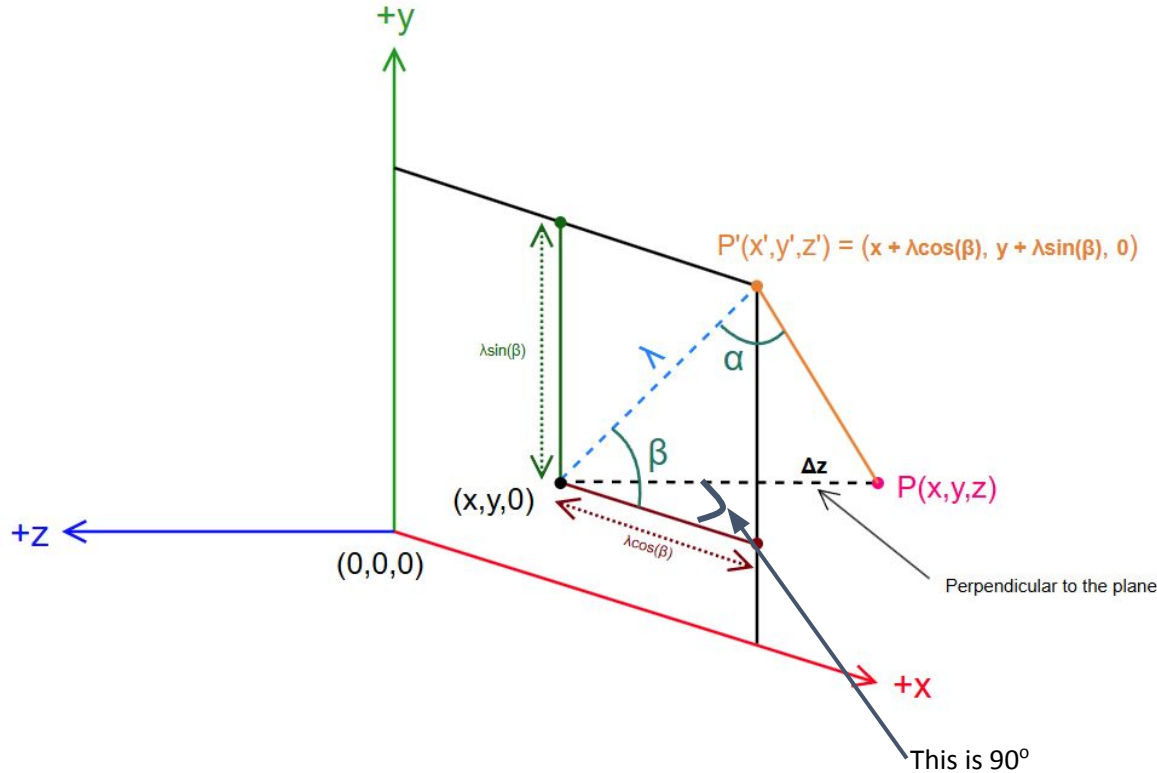
- Depending on the scenario, we need to use a specific projection matrix. A 4x4 homogeneous coordinate matrix since we're in 3D.
- The point to be projected $P(x,y,z)$ will need to be converted into a 4x1 column-matrix, the w/last value will be 1.
- Multiplying the projection matrix and the point column-matrix will give a 4x1 result containing the transformation point.
- The w/last value of result won't necessarily be 1. In this case, we need to divide all values in the result by w to get the actual result called $P'(x',y',z')$.

$$\begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix} \rightarrow \begin{bmatrix} \left(\frac{x' \cdot w}{w} \right) \\ \left(\frac{y' \cdot w}{w} \right) \\ \left(\frac{z' \cdot w}{w} \right) \\ \left(\frac{w}{w} \right) \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

Deriving Parallel Projection Matrix for XY Plane



- The projection rays don't converge, so there are no Center Of Projection (COP)
- There is however, a Direction Of Projection (DOP)
- This DOP is steered by some angles/scaling factors, which control the ultimate projection transformation
- Different angle/scaling factor configuration for Cabinet, Cavalier, Orthographic



P' is the projected point on plane for P

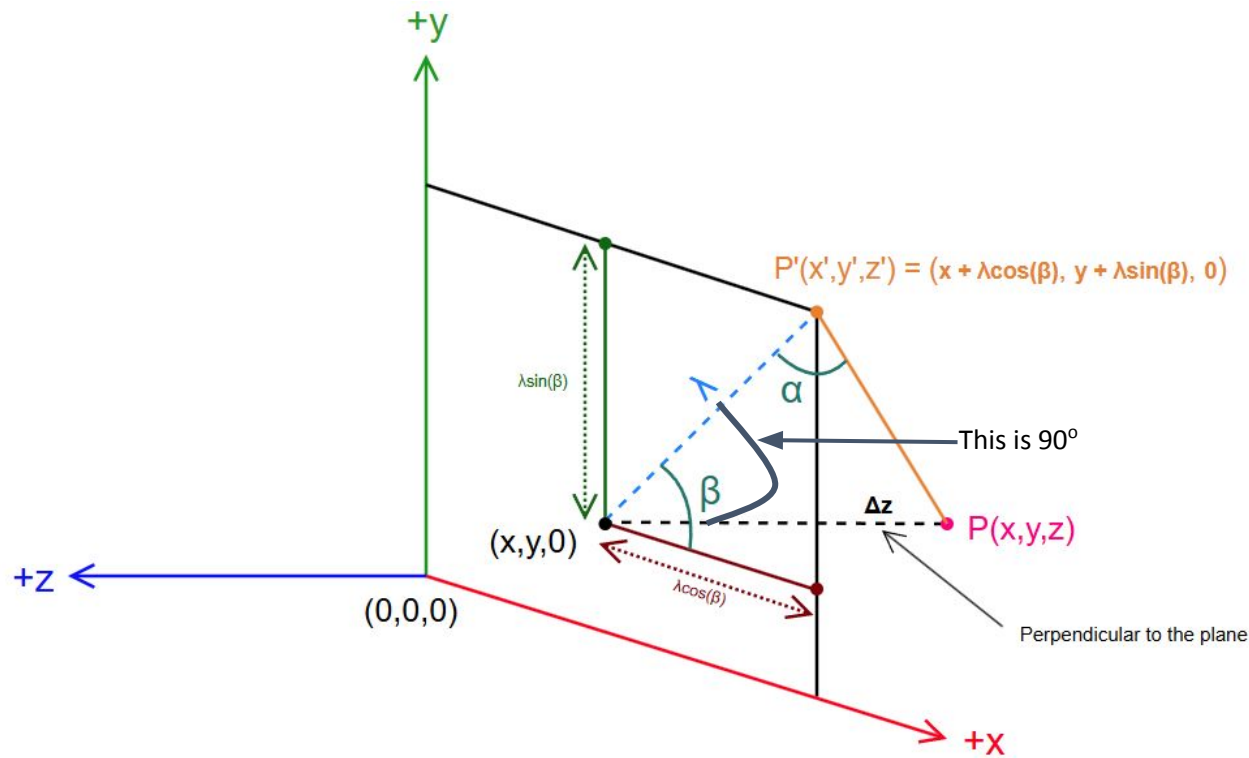
The distance between the plane and P is Δz

$(x, y, 0)$ is an equivalent point for $P(x, y, z)$ put on the projection plane

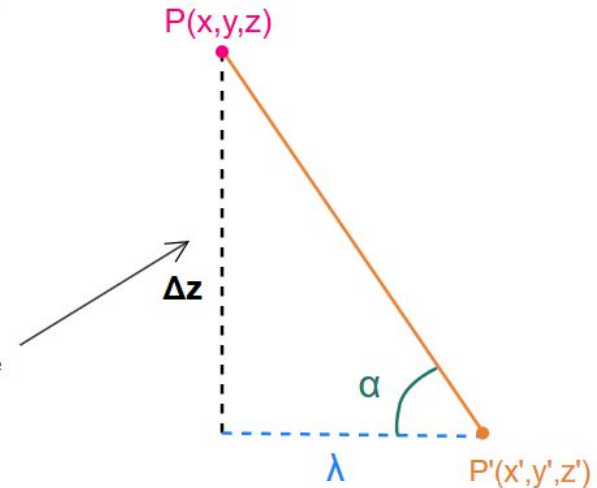
β is inclination angle

$$x' = x + \lambda \cos(\beta) \quad \dots 1(a)$$

$$y' = y + \lambda \sin(\beta) \quad \dots 1(b)$$



α is the angle between the projection plane and projection ray, this is the angle of skewness.



$$\tan(\alpha) = \frac{\Delta z}{\lambda} \Rightarrow \lambda = \frac{\Delta z}{\tan(\alpha)}$$

We will address Δz as just z since:

$$\Delta z = (z - 0) = z$$

From equations 1(a) and 1(b), we can write

$$x' = x + \left(\frac{\cos\beta}{\tan\alpha} \right) z$$

$$y' = y + \left(\frac{\sin\beta}{\tan\alpha} \right) z$$

Since on the projection plane, all z values are 0 (for this particular scenario)

$$z' = 0$$

Rewriting with color coded coefficients

$$x' = (1)x + \left(\frac{\cos\beta}{\tan\alpha} \right) z$$

$$y' = (1)y + \left(\frac{\sin\beta}{\tan\alpha} \right) z$$

$$z' = 0$$

Coefficients Color Coding:
(x = red, y = green, z = blue, leftover = purple)

Strategy: Converting x' , y' , z' equations to a 4x4 matrix

$$x'.w = a_1x + b_1y + c_1z + d_1$$

$$y'.w = a_2x + b_2y + c_2z + d_2$$

$$z'.w = a_3x + b_3y + c_3z + d_3$$

$$w = a_4x + b_4y + c_4z + d_4$$



$$\begin{bmatrix} x'.w \\ y'.w \\ z'.w \\ w \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

We don't have any w equation.

We will consider $w = 1$
(since parallel projection)

$$\begin{aligned} x'.w &= x' \\ y'.w &= y' \\ z'.w &= z' \end{aligned}$$

(Since $w = 1$)

$$\begin{aligned}
 x' &= (1)x + \left(\frac{\cos\beta}{\tan\alpha}\right)z \\
 y' &= (1)y + \left(\frac{\sin\beta}{\tan\alpha}\right)z \\
 z' &= 0 \\
 w &= 1
 \end{aligned}
 \Rightarrow
 \begin{bmatrix}
 1 & 0 & \left(\frac{\cos\beta}{\tan\alpha}\right) & 0 \\
 0 & 1 & \left(\frac{\sin\beta}{\tan\alpha}\right) & 0 \\
 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 1
 \end{bmatrix}
 \times
 \begin{bmatrix}
 x \\
 y \\
 z \\
 1
 \end{bmatrix}
 =
 \begin{bmatrix}
 x'.w \\
 y'.w \\
 z'.w \\
 w
 \end{bmatrix}$$

Since, $w = 1$ (only for this scenario)

$$\begin{bmatrix}
 x'.w \\
 y'.w \\
 z'.w \\
 w
 \end{bmatrix}
 =
 \begin{bmatrix}
 x' \\
 y' \\
 z' \\
 1
 \end{bmatrix}$$

Considering $\Delta z = z$ as 1, we can get this relation:

$$\tan(\alpha) = \frac{1}{\lambda} \quad \Rightarrow \quad \lambda = \frac{1}{\tan(\alpha)}$$

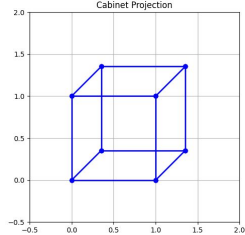
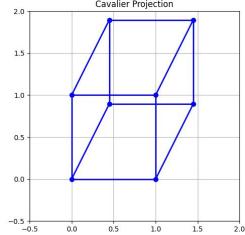
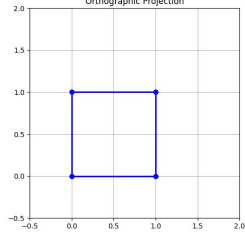
By which, we can get an alternate matrix:

$$\begin{bmatrix} 1 & 0 & \lambda \cos \beta & 0 \\ 0 & 1 & \lambda \sin \beta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix}$$

For the math,

We can either build the matrix using (α, β) or (λ, β)

Configurations for different types of parallel projections

Type	Oblique?	λ	α (deg)	β (deg)	Preview
Cabinet	Yes	0.5	63.4	0-360	 Cabinet Projection
Cavalier	Yes	1	45	0-360	 Cavalier Projection
Orthographic	No	0	90	0-360	 Orthographic Projection

Example: For a point P (10, 20, -40), calculate the projected point P' using Cavalier Projection, orientation angle is 30° for projection plane of XY.

For Cavalier, $\alpha = 45^\circ$. And it is given that, $\beta = 30^\circ$
 The whole calculated matrix with the point multiplied is:

$$\begin{bmatrix} 1 & 0 & 0.866 & 0 \\ 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix} = \begin{bmatrix} -24.64 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

We don't need to divide values by w since w is already 1

P' is (-24.64, 0)



Example: Derive the matrix for orthographic projection

Here, the projection rays will be perpendicular to the projection plane, α will be 90 degrees, making λ as 0, so the value of β won't even be considered. Putting these values into the formula matrix gives:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example: Calculate Orthographic Projection on XY Plane for the point P (d, e, f)

We don't need to know any other value since the orthographic projection matrix is constant, we simply multiply our provided point P (as a homogenous column-matrix) with the matrix.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} d \\ e \\ f \\ 1 \end{bmatrix} = \begin{bmatrix} d \\ e \\ 0 \\ 1 \end{bmatrix}$$

In fact, applying orthographic projection on XY Plane to a point is equivalent to copying the point's x and y values, while setting its z value to 0. [(x, y, z) becomes (x, y, 0)]

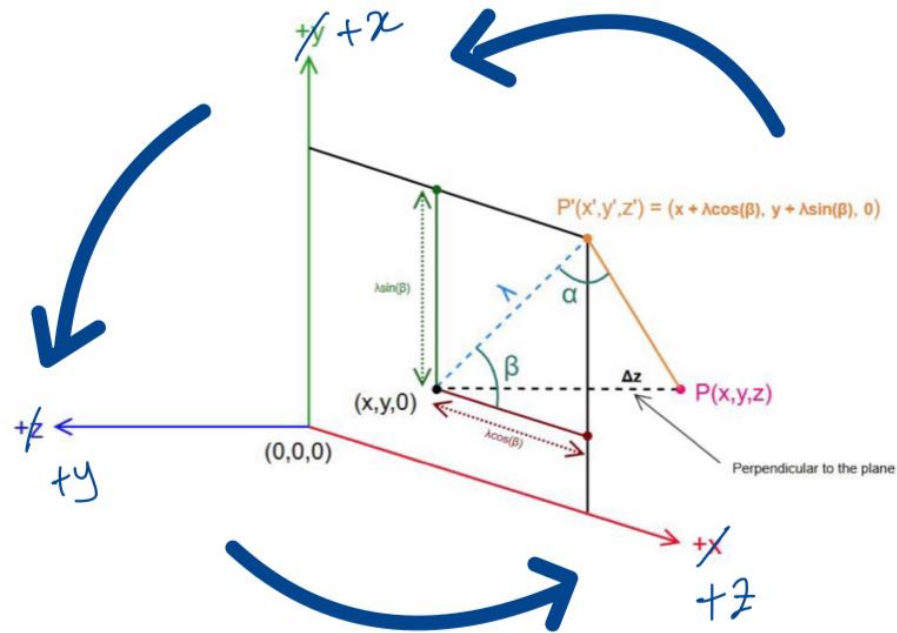
But how can we find the Parallel projection matrix for different projection plane other than XY?

Same way

- Rotate the coordinate system clockwise to get to YZ plane.
- Rotate the coordinate system counterclockwise to get to ZX plane.
- Modify the matrix accordingly.



Projection Plane: ZX / Y = 0



here, $\lambda = \frac{1}{\tan \alpha}$

Actually, $\lambda = \frac{\Delta y}{\tan \alpha}$

$$z' = z + \lambda \cos \beta \times y$$

$$x' = x + \lambda \sin \beta \times y$$

$$\begin{vmatrix} x' \\ y' \\ z' \\ 1 \end{vmatrix} = \begin{vmatrix} 1 & \lambda \sin \beta & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \lambda \cos \beta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \times \begin{vmatrix} x \\ y \\ z \\ 1 \end{vmatrix}$$

Projection Plane: YZ / X = 0

$$y' = y + \lambda \cos \beta \times x$$

$$z' = z + \lambda \sin \beta \times x$$

$$\begin{vmatrix} x' \\ y' \\ z' \\ 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 & 0 \\ \lambda \cos \beta & 1 & 0 & 0 \\ \lambda \sin \beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \times \begin{vmatrix} x \\ y \\ z \\ 1 \end{vmatrix}$$

here, $\lambda = \frac{1}{\tan \alpha}$

Actually, $\lambda = \frac{\Delta x}{\tan \alpha}$

